Detecting AI Hallucination Patterns in Language Model Responses

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STAT 4640 - Final Project Report

***Abstract*—Large language models have become widely used for generating natural language responses, but they are also known to produce hallucinations, where outputs appear convincing but are factually incorrect. This project analyzes hallucination behavior using the TruthfulQA dataset, which contains questions and both correct and incorrect responses across multiple domains.**

A dataset was constructed by combining truthful responses and hallucinated responses into a binary classification problem. Features such as question length, answer length, and sentence count were extracted and used as predictors. A Bayesian logistic regression model was then fit using the `brms` package in R to estimate the probability of hallucination.

The results show that sentence count has a positive relationship with hallucination probability, while answer length has a slight negative effect. Question length does not appear to have a significant influence. Posterior predictive checks indicate that the model captures the general structure of the data. These findings provide insight into how response complexity affects hallucination behavior and demonstrate how Bayesian methods can be applied to analyze modern AI systems.

I. Introduction

Large language models are capable of generating fluent and coherent responses across a wide range of topics. However, they are also prone to producing hallucinations, where the generated output is incorrect but still appears believable. This presents challenges in applications where reliability is important, such as education, research, and decision-making.

In this project, hallucination behavior is analyzed using the TruthfulQA dataset. This dataset contains 817 questions across different categories, including health, science, and politics. The goal is to understand what characteristics of prompts and responses are associated with hallucinated outputs.

The response variable is defined as:

- $y_i = 1$ if the response is hallucinated
- $y_i = 0$ if the response is truthful

The main research question is:

What characteristics of prompts and responses increase the probability of hallucination?

II. Exploratory Data Analysis

Before fitting the model, the dataset was explored to understand its structure and key features.

The distribution of question length was examined to understand how complex the prompts are.

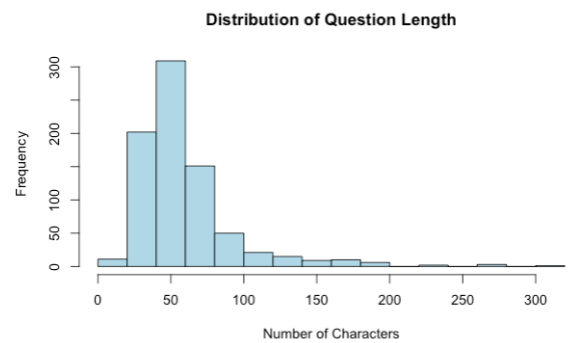

Fig. 1. Distribution of question length in the TruthfulQA dataset. Most questions fall between roughly 30 and 80 characters, though a small number of longer questions extend beyond 200 characters.

Most questions fall within a moderate length range, with a small number of longer questions acting as outliers. This suggests that the dataset includes a mix of simple and more detailed prompts.

The dataset was also examined across categories to understand coverage.

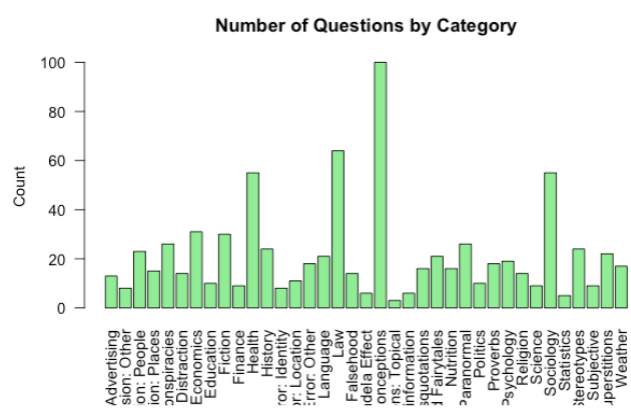

Fig. 2. Number of questions across different categories in the TruthfulQA dataset.

The results show that the dataset spans multiple domains, which is useful for analyzing hallucination behavior across different topics.

These exploratory steps help justify the choice of predictors used in the model.

III. Bayesian Model

To analyze hallucination behavior, a Bayesian logistic regression model was used. The outcome variable is modeled as:

$$Y_i \sim \text{Bernoulli}(p_i)$$

The probability of hallucination is modeled using:

$$\text{logit}(p_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i}$$

where:

- X_{1i} = question length
- X_{2i} = answer length
- X_{3i} = sentence count

Weakly informative priors were used:

$$\beta_j \sim \mathcal{N}(0, 10^2)$$

This model allows estimation of how each predictor influences the log-odds of hallucination.

IV. Model Fitting

The dataset was constructed by combining truthful responses from the *Best Answer* column and hallucinated responses from the *Best Incorrect Answer* column. This resulted in a dataset of 1580 observations.

The model was fit using the `brms` package in R with 4 chains and 2000 iterations per chain.

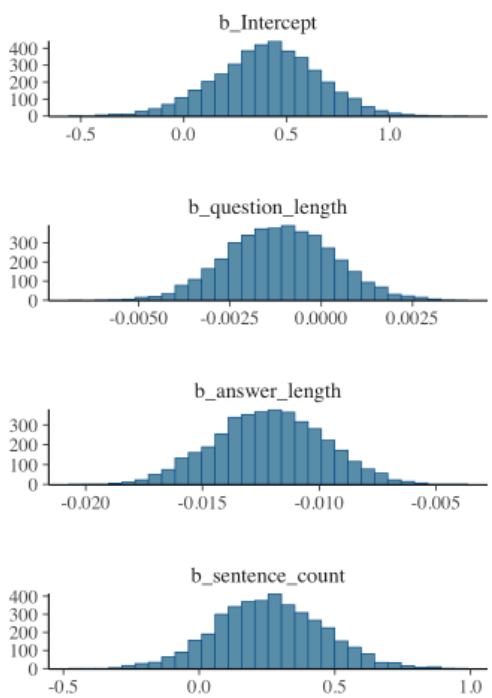

Fig. 3. Posterior density distributions of the model coefficients. The distributions are smooth and unimodal, suggesting stable parameter estimates. The results show that answer length has a negative effect, while sentence count has a positive effect on hallucination probability.

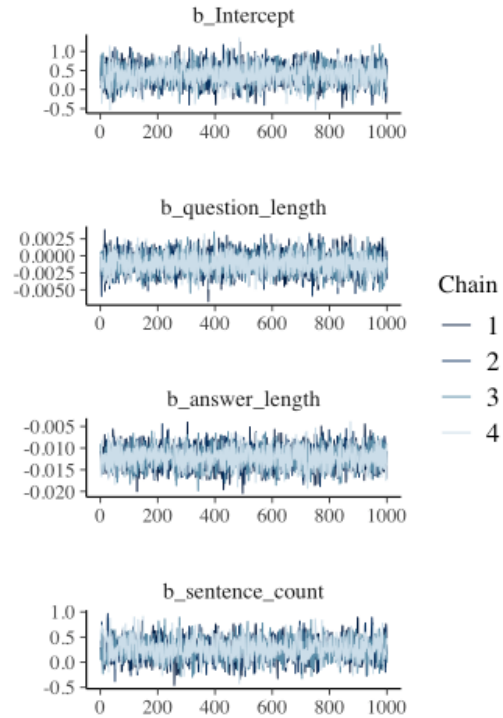

Fig. 4. Trace plots of the regression coefficients across all chains. The chains show good mixing with no visible trends or drift, indicating that the Markov Chain Monte Carlo sampling converged successfully.

Trace plots were used to assess convergence, and all $\hat{R}$ values were approximately 1. This indicates that the model converged successfully and the posterior estimates are reliable.

V. Posterior Analysis

The posterior results provide insight into how different predictors influence hallucination probability.

Question length has a coefficient close to zero, indicating that it does not have a noticeable effect on hallucination.

Answer length has a small negative coefficient (-0.01), suggesting that longer answers are slightly less likely to be hallucinated. This may indicate that more detailed responses are more likely to remain accurate.

Sentence count has a positive coefficient (0.26), indicating that responses with more sentences are more likely to be hallucinated. This suggests that more complex responses may introduce a higher chance of incorrect information.

Overall, these results suggest that the structure and complexity of responses play a role in hallucination behavior.

VI. Model Assessment

Posterior predictive checks were used to evaluate how well the model fits the data.

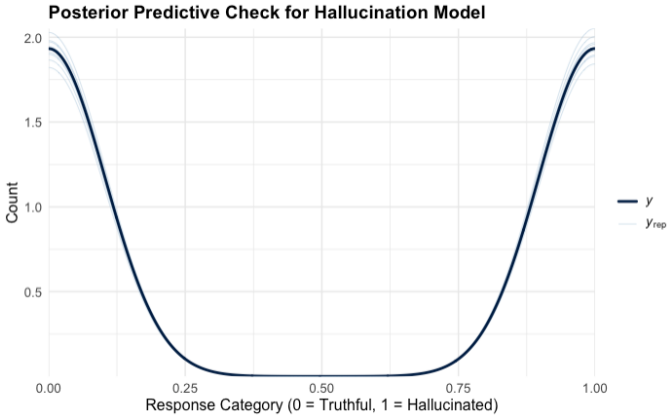

Fig. 5. Posterior predictive check comparing the observed data (y) with simulated data (y_{rep}) generated from the fitted model. The close overlap between the observed and predicted distributions indicates that the model is able to capture the overall pattern of hallucinated and truthful responses, suggesting a reasonable model fit.

The results show that the simulated data from the model generally follows the same pattern as the observed data. While there are some differences, the model captures the overall structure reasonably well.

This indicates that the model is appropriate for the data, although there may still be room for improvement.

VII. Conclusion

This project analyzed hallucination behavior in language model responses using a Bayesian approach. The results show that sentence complexity has a noticeable effect on hallucination probability, while question length has little impact.

There are some limitations to this analysis. The hallucination variable was constructed using available dataset columns, which may not perfectly represent all cases. In addition, only a small number of predictors were used.

Future work could include using more advanced text features, improving labeling methods, or applying more complex models.

References

[1] **DATASET REPOSITORY LINK:** S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring How Models Mimic Human Falsehoods," 2021. [Online]. Available: https://github.com/sylinrl/TruthfulQA

Appendix A
Appendix: R Code

```
# Loading the needed libraries
library(brms)
library(tidyverse)
library(stringr)

# Loading the dataset
data <- read.csv("TruthfulQA.csv")

# Adding the features
data$question_length <- nchar(data$Question)
data$answer_length <- nchar(data$Best.Answer)
data$sentence_count <- sapply(
  str_split(data$Best.Answer, "\\."),
  function(x) length(x)
)

# Constructing the hallucination parts

# Truthful responses
truth_data <- data.frame(
  text = data$Best.Answer,
  label = 0
)

# Hallucinated responses
halluc_data <- data.frame(
  text = data$Best.Incorrect.Answer,
  label = 1
)

# Combining the datasets
combined_data <- rbind(truth_data, halluc_data)

# Final features (final dataset)
combined_data$answer_length <- nchar(combined_data$text)

combined_data$sentence_count <- sapply(
  str_split(combined_data$text, "\\."),
  function(x) length(x)
)

combined_data$question_length <- rep(
  nchar(data$Question),
  2
)

# Final response variable
combined_data$hallucination <- combined_data$label

# Fitting the bayesian logistic regression model
fit <- brm(
  hallucination ~ question_length + answer_length + sentence_co
  data = combined_data,
  family = bernoulli(link = "logit"),
  chains = 4,
  iter = 2000,
  cores = 4
)

# Results

summary(fit)

# Convergence diagnostics
plot(fit)
rhat(fit)

# Posterior predictive check
pp_check(fit)
```